Learning with Mala

Mth622

Quiz No.4

Important McQs

Final Term Preparation 2025

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Question # 1 of 10 (Start time: 12:25:09 PM, 04 July 2025)

Total Marks: 1

The moment of inertia of a ring of mass 1kg and radius 0.9m about an axis passing through its center will be _____kgm^2.

Select the correct option

Reload Math Equations

- 1.8
- 0.81
- 0.45
- 0.18

$$I = MR^2$$

Where:

- $M = 1, \text{kg}$ (mass)
- $R = 0.9, \text{m}$ (radius)

Now calculate:

$$I = 1 \times (0.9)^2 = 1 \times 0.81 = 0.81\,\text{kg} \cdot \text{m}^2$$

sec(s)

MTH622 - Vectors and Classical Mechanics (Quiz 4)

Quiz Start Time: 12:25 PM, 04 July 2025

Question # 1 of 10 (Start time: 12:25:09 PM, 04 July 2025)

Total Marks: 1

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Select the correct option

Reload Math Equations

- 1.8
- 0.81
- 0.45
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Time Left sec(s)

MTH622 - Vectors and Classical Mechanics (Quiz 4)

Quiz Start Time: 12:25 PM, 04 July 2025

Question # 2 of 10 (Start time: 12:26:16 PM, 04 July 2025)

Total Marks: 1

Let 2kg be the mass and 0.5m be the radius of a right circular cone. Its moment of inertia about the axis of symmetry will be ____kgm^2.

Select the correct option

Reload Math Equations

- 0.15
- 0.42
- 0.50
- 0.35

$$I = \frac{3}{10} M R^2$$

Where:

- $M = 2, \text{kg}$ (mass)
- $R = 0.5, \text{m}$ (radius)

Plug in the values:

$$I = \frac{3}{10} \times 2 \times (0.5)^2 = \frac{3}{10} \times 2 \times 0.25 = \frac{3}{10} \times 0.5 = 0.15\,\text{kg} \cdot \text{m}^2$$

sec(s)

MTH622 - Vectors and Classical Mechanics (Quiz 4)

Quiz Start Time: 12:25 PM, 04 July 2025

Question # 3 of 10 (Start time: 12:26:53 PM, 04 July 2025)

Total Marks: 1

A solid metal sphere having the moment of inertia of $0.03 kgm^2$ is rotating with an angular speed of 13.8 rad/s. Its angular momentum would be _____kgm^2s^{-1}.

Select the correct option

Reload Math Equations

- 0.440
- 0.354
- 0.414
- 0.204

$$L = I \cdot \omega$$

Where:

- $I = 0.03, \mathrm{kg} \cdot \mathrm{m}^2$ (moment of inertia)
- $\omega = 13.8, \mathrm{rad/s}$ (angular speed)

Calculate:

$$L = 0.03 \times 13.8 = 0.414 \, \mathrm{kg} \cdot \mathrm{m}^2/\mathrm{s}$$

sec(s)

MTH622 - Vectors and Classical Mechanics (Quiz 4)

Quiz Start Time: 12:25 PM, 04 July 2025

Question # 4 of 10 (Start time: 12:27:32 PM, 04 July 2025)

Total Marks: 1

If no external force is acting on the system of two colliding bodies, then by the principle of conservation of linear momentum which of the following statement is correct?

Select the correct option

Reload Math Equations

- Total momentum before the collision is less than the total momentum after the collision.
- Total momentum before the collision is greater than the total momentum after the collision.
- Total momentum before the collision is equal to the total momentum after the collision.
- Total momentum before the collision is reciprocal of the total momentum after the collision.

sec(s)

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Quiz Start Time: 12:25 PM, 04 July 2025

Question # 6 of 10 (Start time: 12:29:13 PM, 04 July 2025)

Total Marks: 1

A solid metal sphere having the moment of inertia of $0.44 kgm^2$ is rotating with an angular speed of 2 rad/s.Its rotational kinetic energy would be _____ J.

Select the correct option

Reload Math Equations

- 0.88
- 0.76
- 0.48
- 0.66

$$K.E. = \frac{1}{2} I \omega^2$$

Where:

- $I = 0.44, \text{kg} \cdot \text{m}^2$ (moment of inertia)
- $\omega = 2, \text{rad/s}$ (angular speed)

Plug in the values:

$$K.E. = \frac{1}{2} \times 0.44 \times (2)^2 = 0.5 \times 0.44 \times 4 = 0.88\,\text{J}$$

sec(s)

MTH622 - Vectors and Classical Mechanics (Quiz 4)

Quiz Start Time: 12:25 PM, 04 July 2025

Question # 8 of 10 (Start time: 12:30:36 PM, 04 July 2025)

Total Marks: 1

Let P be any point on the ellipsoid of inertia and 0.25kgm^2 be the moment of inertia about any line OP of the rigid body. The length of line OP will be _____m.

Select the correct option

Reload Math Equations

- (●) 2
- ( ) 4
- ( ) 1
- ( ) 5

Click to Save Answer & Move to Next Question

sec(s)

MTH622 - Vectors and Classical Mechanics (Quiz 4)

Quiz Start Time: 12:25 PM, 04 July 2025

Question # 9 of 10 (Start time: 12:32:10 PM, 04 July 2025)

Total Marks: 1

What happens to angular velocity when the moment of inertia of an isolated system is doubled to keep the angular momentum constant?

Select the correct option

Reload Math Equations

- Remains same.
- It becomes zero.
- It is halved.
- It doubles.

sec(s)

MTH622 - Vectors and Classical Mechanics (Quiz 4)

Quiz Start Time: 12:25 PM, 04 July 2025

Question # 10 of 10 (Start time: 12:32:51 PM, 04 July 2025)

Total Marks: 1

If the center of mass is at origin then the coordinate axes (xyz axes) and the principal axes of the rigid body are.....

Select the correct option

Reload Math Equations

- coincident
- perpendicular

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